

Now let's look at the operations in `first_gsl.c`, going by the comments with numbers. At first glance, most of the code doesn't seem to be complex; we just call a function to get what we want.

GSL header files: By default, all the header files (that were installed by the *make install* earlier) are stored in `/usr/local/include/gsl`. If you haven't changed the default location, you can just compile using the following command:

```
$ gcc -Wall -c first_gsl.c
```

Regarding the linking part: if you have installed GSL in its default location, the library is installed as a single file, `libgsl.a`, in `/usr/local/lib`. So, again, you can just compile with the following command:

```
$ gcc first_gsl.o -lgsl -lgslcblas -ls
```

GSL data types: GSL data structures are nothing but custom data types required by the specific type of application. In the above example, we needed a complex number, which has real and imaginary values. So the GSL structure was defined as:

```
typedef struct
{
    double dat[2];
} gsl_complex;
```

Briefly, the `gsl_complex` struct contains an array that will store the real and imaginary types. We will discuss data types for specific applications in the next article.

References

1. M. Galassi et al, GNU Scientific Library Reference Manual (Edition 1.14, for GSL Version 1.14), ISBN 0954612078
2. <http://www.gnu.org/software/gsl/>

By: Amol S Takkalki

The author is a FOSS enthusiast, whose interests include data mining, neural networks, process automation and high-performance/parallel computing. His hobbies are programming, photography, Linux and music. You can contact him at takkalki.amol@gmail.com. He would like to say, "Open source rocks!"

Knowing the data-type structure will help you to use it more efficiently (as demonstrated in Number 4, below).

Using GSL functions: Like I said, you should know the data type that you will be using, and its format; this will be useful in passing and handling returned parameters. All the functions that we will use are explained in detail in the documentation, which you can download from the GSL homepage.

Using GSL parameters: Again, knowing the format/definition of GSL variables enables you to extract the appropriate terms from it. Looking at the type definition told me that the struct contained an array of size 2, named `dat`. Thus, I could print the real part and the imaginary part by accessing the member of the struct, and the appropriate array index.

I have introduced some basic information that should start off people who're new to GSL. Personally, I have been using GSL for about 1.5 years, and have found that most of my work has been quite a bit easier and quicker—it is possible to concentrate more on the bigger picture, instead of worrying about smaller details.

If you wish to contribute to GSL, you can find the ongoing plans in the `TODO` file in the archive-extracted directory. It includes all resolved issues, and issues to be resolved in the next release. It also gives a list of all routines that are to be added in the next release.

In forthcoming articles, I will touch on fields with their data types, and the different functions involved. See you soon! 

IPSR launches RHCVA

(Red Hat Certified Virtualization Administrator)



Get trained by faculty team headed by RHCVA Certified professionals

Red Hat Certification Exams EVERY WEEK!

Advantage RHCVA

- There are no prerequisites for this certification. Any candidate who has knowledge in System Administration (Windows / Linux / Unix / Sun Solaris) can opt for this certification
- The certification consists of just 1 module

RHCE RHCS RHCD RHCA

IPSR, The Best in Red Hat Certifications

Awards

- Winner of Red Hat Asia Pacific International Award 08 & 09
- Winner of Red Hat GLS National Awards 04, 05, 06, 07 & 08

Faculty & Infrastructure

- 15 member faculty crew led by RHCE, RHCVA, RHCS, RHCD, RHCA certified professionals
- Global standard infrastructure offered at our International Centre located in Kochi

Achievements

- Produced 2500+ RHCEs & 24 RHCSSs as of June 2010, the highest in the world by a single partner
- Produced the youngest RHCE (11 years old) & Youngest RHCS (13 years old)

ipser solutions ltd
redefining excellence

Corporate Office : Kottayam
Branches: Kochi North Kochi South, Kozhikode
Tel : 0484 - 2344560, 0481 - 2561410 / 20
Mob. 9447294635, 9447169776
www.ipser.org • training@ipserolutions.com